Roadside Accidents In India And Image Preprocessing For Autonomous Cars

Yash Mishra, Vijay Kumawat, Jasmin T. Jose

Abstract: Roadside accidents are one of the leading cause of deaths in India. Though car accidents are preventable, number of causalities due to car accidents have only seen an increase over the years. This paper looks at one of the promising solutions to this problem – autonomous cars. While self-driving cars dominating the Indian roads is still a distant future, it is important to understand and improve the image preprocessing required in a self-driving car before they are omnipresent in our society. This paper interprets the released government data on roadside accidents in India and studies different types of noise with varying intensity to help choose the most appropriate smoothing filter in spatial domain. The paper then focuses on another important aspect of pre-processing, edge detection and compare the prevalent edge-detection techniques with one another.

Key words: Average Filter, Canny edge detection , Edge Detection Gaussian Filter, Image Processing, Noise Reduction, Roadside Accidents, Smoothing Filter

1.INTRODUCTION

This paper looks into the roadside accidents in India and the image pre-processing involved in autonomous cars. India has constantly seen the number of road accidents increase over the years with over four lakh accidents in the year 2018. [1] The same report also ranked India at the top in the list of countries with the highest number of road accident related deaths, even ahead of China which has a higher population than India. India contributed to almost 11% of deaths caused to roadside accidents in the world. This paper is divided into three sections. In the first section, government data made available to the public is visualized and analyzed to gain insight into the situations and understand where each states and union territories stand in terms of road safety for the citizens. In the second section, this paper compares the efficiency of Gaussian and Average Smoothing Filters to remove three types of Noise: Average, Speckle and Impulse (Salt and Pepper Noise) and how parameters involved in the filters affect the result. Mean Square Error and Signal to Noise Ratio (SNR) were chosen as the two parameters to judge the smooth images achieved. We compare two different smoothing filters, Average and Gaussian, for JPEG images of size 667 x 500, in which three types of noises – Gaussian Noise, Speckle Noise and Salt and Pepper Noise were added with varying intensity and then their performance was measured using Mean Square Error and Signal to Noise Ratio as parameter. In the third section, a photograph is taken and its edges are manually drawn to be used as ground truth to assess the performance of various edge detections methods using Pratt's Figure of Merit.

- Yash Mishra is currently pursuing Bachelor's degree program in computer science engineering in Vellore Institute of Technology, Vellore, TN, INDIA, E-mail: yashmishra12@hotmail.com
- Yash Mishra is currently pursuing Bachelor's degree program in computer science engineering in Vellore Institute of Technology, Vellore, TN, INDIA, E-mail: vijaykumawat256@gmail.com
- Jasmin T Jose is currently an Assistant Professor in Vellore Institute of Technology, Vellore, TN, INDIA, E-mail: jasminlijo@vit.ac.in

2.DATA VISUALISATION

Figure 1 from the first dataset [4] shows that even though the number of road-accidents have decreased over the years, the number people killed have steady increased, thereby indicating towards an increased number of fatal accidents. It also shows that the government has worked on the roadways connectivity and increased the total length of roads in India, and the number of accidents per thousand kilometres of road has consistently decreased but the number of people killed have only gone worse, with more people dying with fewer accidents every year.

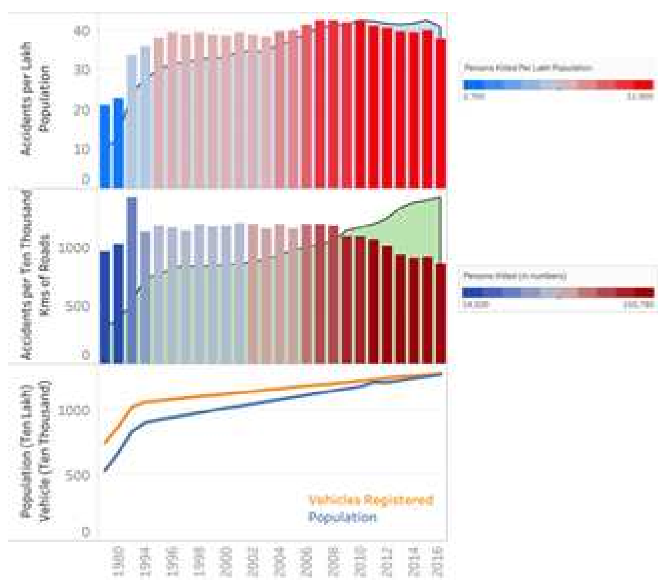

Figure 1. Accidents in India over the years with respect to increasing road length and population

Other datasets [5][6] reveal that almost three quarters of the accidents and almost seventy percent of the total deaths in accidents occur on sunny and clear days, indicating towards the minor role weather condition plays in accident. And the time in the evening, between 3PM to 6PM see the highest number of car accidents, which might also be due the number of cars present on road then.

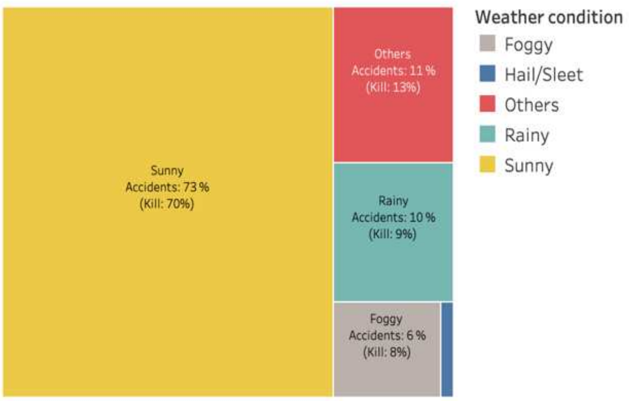

Figure 2. *Accidents with respect to weather condition*

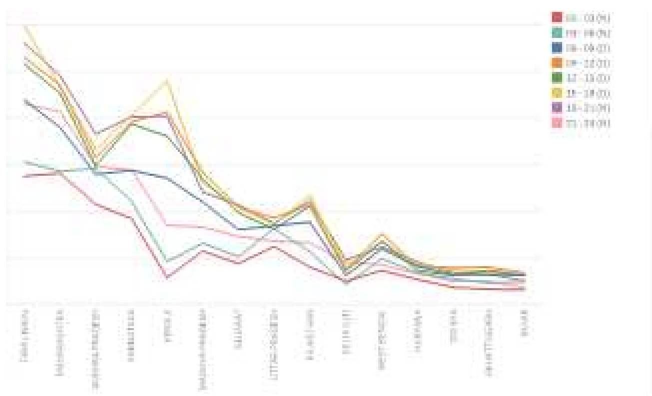

Figure 3. *Number of accidents with respect to time in major states*

As shown in Figure 4, another dataset [7] shows that while Maharashtra had the highest number of car accidents, the state of Uttar Pradesh and Andhra Pradesh saw the most deaths on road.

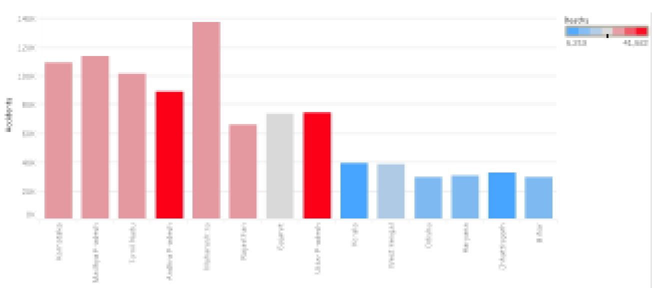

Figure 4. *Number of accidents in states of India with the number of deaths.*

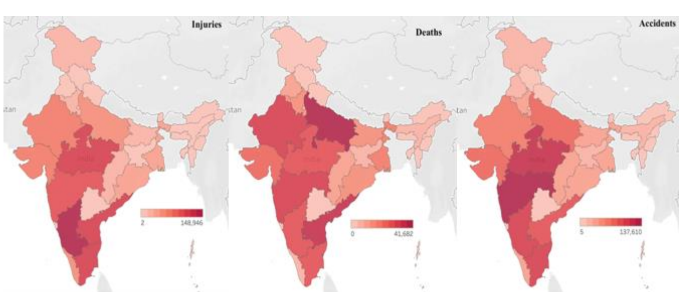

***Figure 5**. (Left to Right) Number of Injuries, Deaths and Accidents 2006-2016*

3. SMOOTHING FILTER IN SPATIAL DOMAIN

Smoothing is a process of eliminating noise and other similar aberrations from an image. Image smoothing is an integral and crucial component of image enhancement, which can remove noise in images. A smoothing algorithm is judged on its capability to remove noise and preserve edges of the image. The difference in the algorithm of each individual smoothing filter results in a different output for each noise model which makes the choice of the filters very important. Smoothing can be done either in the spatial domain or in the frequency domain [2], but Spatial domain filtering is simpler and faster than its counterpart [3] and since even a millisecond could be the difference between injury and fatality, this paper focuses on spatial domain filters. 8100 JPEG images were resized to a dimension of 667x500 pixel. Images were collected using google chrome extension which allow batch download of Google Images. The MSE and SNR values found is the average of all images taken. The parameter for average filter is filter size (n x n) and for Gaussian filter it is sigma value. The parameters were changed gradually keeping the noise percentage constant and the respective SNR and MSE values were noted. Images were tested with six different intensity levels of noise, namely: 5%, 10%, 20%, 40%, 60%, 85%.

3.1. MEAN SQUARE ERROR

The Mean Square Error (MSE) between two images – X(i,j) and Y(i,j) is defined as:

$$\boldsymbol{MSE} = \frac{1}{(N \times M)} \sum_{i=0}^{N-1} \sum_{j=0}^{M-1} [\mathrm{X(i,j)} - \mathrm{Y(i,j)}]^2$$

Where N and M are the dimensions of both the images respectively [8]. One problem with Mean Square Error is that it depends strongly on the image intensity scaling [9].

3.2. SIGNAL TO NOISE RATIO

Signal to Noise Ratio (SNR) overcomes the problem of MSE [10] discussed earlier and it is measured in decibels (db.)

$$\boldsymbol{SNR} = 20\ log_{10} \frac{signal}{RMS\ noise}\ dB$$

3.3. GAUSSIAN FILTER

Gaussian filtering g is used to blur images and remove noise and add detail. In one dimension, the Gaussian function is [11]:

$$G(x) = \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{-\frac{x^2}{2\sigma^2}}$$

Where σ is the standard deviation of the distribution. The distribution is assumed to have a mean of 0. In probabilistic terms, it describes 100% of the possible values of any given space when varying from negative to positive values.

3.4. AVERAGE FILTER

Average filtering is a simple, intuitive and easy to implement method of smoothing images, i.e. reducing the amount of intensity variation between one pixel and the next. It is often used to reduce noise in images. The idea of average filtering is simply to replace each pixel value in an image with the average value of its neighbors, including itself. This has the effect of eliminating pixel values which are unrepresentative of their surroundings [12].

3.5. RESULTS

The x-axis of the graphs is titled as ‘Filter Size’ which refers to the size of Average filter taken, that is, the value of N in the N*N filter and the value of sigma for the Gaussian Filter. Gaussian filter has an additional parameter-sigma. Higher the sigma value, higher will be the significance of central values of the filter while smoothing. As a thumb rule, higher the value when Signal to Noise Ratio is taken in Y-axis, the better the smoothing and lower the value for Mean Square Error, the better is the smoothing filter performance.

3.5.1. GAUSSIAN NOISE.

Gaussian noise is a type of statistical noise in which the amplitude of the noise follows that of a Gaussian distribution. [13] Gaussian Noise occurs as the probability density function of the normal distribution. Thus, Gaussian Noise represents the frequency spectrum that has a bell-shaped curve. Figure 7 suggests that for Gaussian at all noise intensities, Gaussian Filter performs better than Average filter for all filter sizes according to SNR but if we consider MSE as a parameter, which tells us about the pixel intensity difference, then for very high noise intensity, Average smoothing Filter performs slightly better than Gaussian filter but the difference is minuscule according to Figure 6. Hence, we can conclusively say that Gaussian Filter is the best option to remove Gaussian noise.

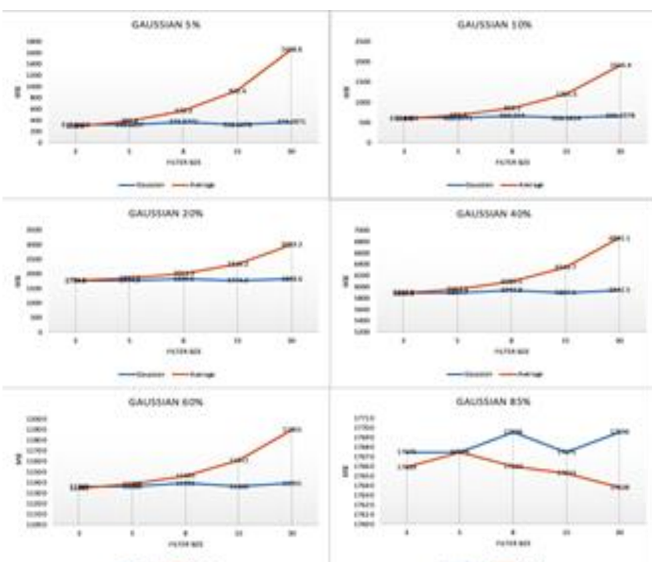

Figure 6. *MSE value for varying intensity of Gaussian noise for Average and Gaussian Filter*

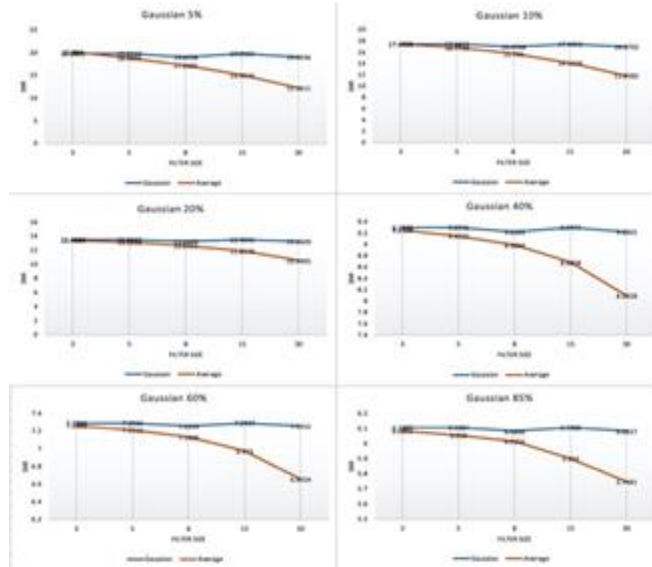

Figure 7*. SNR value for varying intensity of Gaussian noise for Average and Gaussian Filter*

3.5.2. SPECKLE NOISE.

Speckle noise is a granular noise that increases the mean grey level of a local area in an image. [14] This type of noise makes it difficult for image recognition and interpretation. In this noise type, the sample mean and variance of a single pixel is proportional to that of the mean and variance of the local area that is centered on that pixel. It is a deterministic, random and consists of an interference pattern. For Speckle Noise, things are not so simple and results change with filter size. At 5% speckle noise, according to MSE (Figure 8), Gaussian filter performs uniformly good irrespective of the sigma value but for higher noise intensities, Average smoothing Filter performs better than Gaussian filter for most of the filter sizes and Gaussian Filter starts performing better only at high filter sizes. Similar trend is observed if the parameter chosen is SNR (Figure 9) where it is observed that Gaussian Filter is consistent with its performance but Average Filter is a better choice for higher noise intensity.

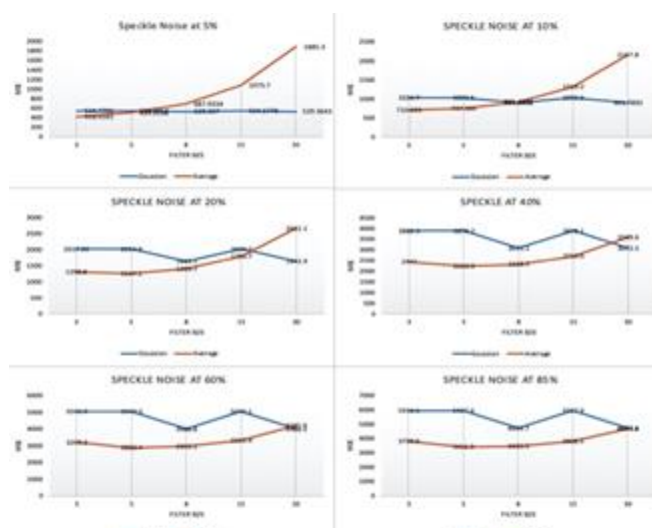

Figure 8*. MSE value for varying intensity of Speckle noise for Average and Gaussian Filter*

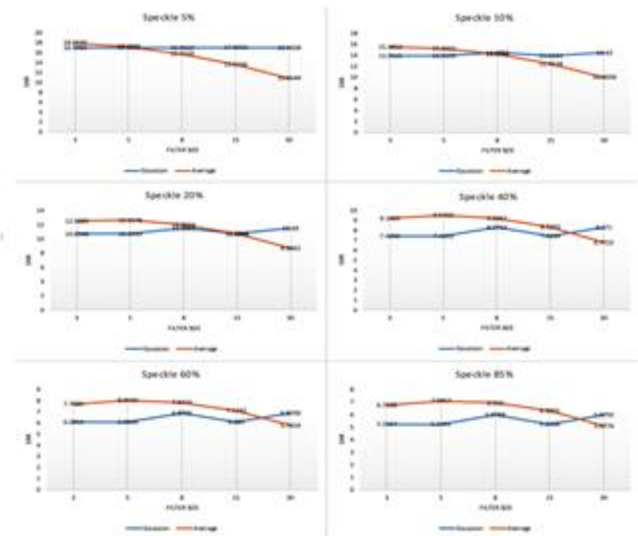

Figure 9. *SNR value for varying intensity of Speckle noise for Average and Gaussian Filter*

3.5.3. SALT AND PEPPER NOISE

An image containing salt-and pepper noise will have dark pixels in bright regions and bright pixels in dark regions. Salt and pepper noise are predominantly found in digital transmission and storage [15]. This noise type can be caused by dead pixels, analog to digital converter errors and bit errors in transmission. Salt and Pepper noise removal technique is heavily governed by filter size. For lower noise intensities, Average filter performs better than Gaussian filter for lower filter sizes and at higher filter size, Gaussian performs better if difference between pixel intensities (MSE) is considered (Figure 10). At higher noise intensities, Average Filter performs better than Gaussian filter for all filter size. The same trend is followed if our parameter changes to SNR.

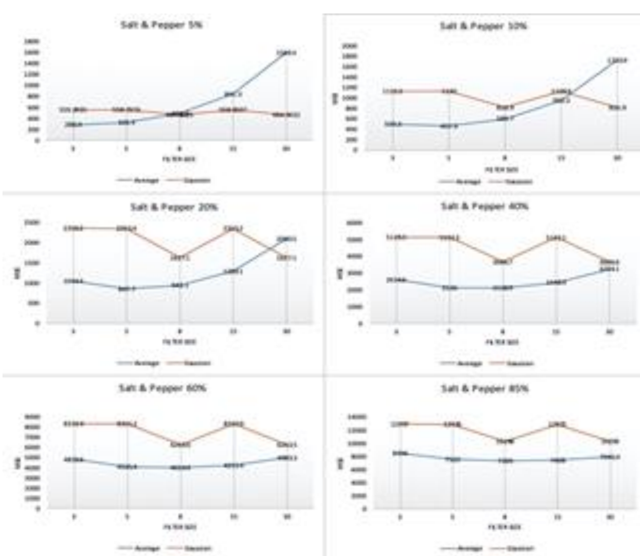

Figure 10. *MSE value for varying intensity of Salt & Pepper noise for Average and Gaussian Filter*

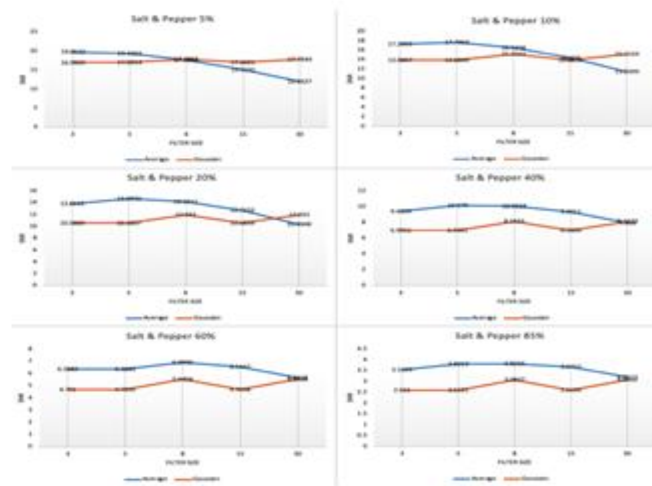

Figure 11. *SNR value for varying intensity of Salt & Pepper noise for Average and Gaussian Filter*

4. EDGE DETECTION

Edge detection is one of the most important steps in computer vision and is generally performed after noise removal and converting the given image into grayscale. Any edge detection method aims at facilitating image analysis by, on one hand, reducing the amount of data that is to be processed, while on another, also preserving useful object structural information and object boundary. [16]

4.1. CANNY EDGE DETECTOR

One of the common methods of edge detection is called Canny Edge Detection introduced by John F Canny, University of California, Berkeley in 1986. It basically uses multiple steps including Gaussian filters, intensity gradient changes to determine edges [16]. Steps involved in Canny [17]:

Step 1. Smooth the given image f(m,n) with a Gaussian Filter.

$$g(m,n) = G_\sigma(m,n) * f(m,n)$$

$$\text{Where, } G_\sigma(m,n) = \frac{1}{\sqrt{2\pi\sigma^2}} exp(\frac{-m^2-n^2}{2\sigma^2})$$

Step 2. Compute the gradient of g(m,n) using any of the existing methods like Robert, Prewitt, Sobel

$$M(m,n) = \sqrt{g_m^2(m,n) + g_n^2(m,n)}$$

$$\theta(m,n) = \tan^{-1}[\frac{g_n(m,n)}{g_m(m,n)}]$$

Step 3. Calculate the threshold M

$$M_T(m,n) = \begin{cases} M(m,n) \text{ } if \text{ } M(m,n) > T \\ 0 \text{ } otherwise \end{cases}$$

T is chosen so that the maximum edge elements are preserved and maximum noise is suppressed.

Step 4. Thin the edges which might have been broadened in Step 1

$$M_T(m,n) = \begin{cases} M(m,n) \text{ if } *condition \\ 0 \text{ otherwise} \end{cases}$$

*condition - Non-zero $M_T(m,n)$ is greater than its two neighbours along the direction of θ(m,n)

Step 5. Threshold the previous result by two different thresholds τ_1 and τ_2 (where $\tau_1 < \tau_2$) to obtain two binary images T_1 and T_2.

Step 6. Link edge segments in T_1 to form continuous edges. To do so, trace each segment in T_2 to its end and then search its neighbours in T_1 to find any edge segment in T1 to bridge the gap until reaching another edge segment in T_2.

4.2. ROBERT EDGE DETECTOR

Robert Edge detection creates the gradient of the image using Robert Cross

+1	0
0	-1

Gx

0	+1
-1	0

Gy

4.3. SOBEL EDGE DETECTOR

The following kernel is used by Sobel to find the edge of the image.:

-1	0	+1
-2	0	+2
-1	0	+1

Gx

+1	+2	+1
0	0	0
-1	-2	-1

Gy

4.4. PREWITT EDGE DETECTOR

The following kernel is used by Prewitt to find the edge when the kernel size is set to 3x3:

+1	0	-1
+1	0	-1
+1	0	-1

Gx

-1	-1	-1
0	0	0
+1	+1	+1

Gy

4.5. HOLISTICALLY-NESTED EDGE DETECTION

Canny Edge Detection algorithm is more than three decades old but still is the most widely used edge detection algorithm currently. Many have proposed new algorithms which claim to be better and robust. Holistically-Nested Edge Detection (HED) uses convolutional neural network and deeply supervised nets to detect edges [18]. HED is guided by deep supervision to learn the hierarchical representation in the image which helps it to distinguish object boundaries from edges in the image.

4.6. PRATT'S FIGURE OF MERIT

The Pratt's Figure of Merit (PFOM) is a gives a numerical comparison among various edge detection algorithm. Its value ranges from 0 to 1, where 1 shows maximum match between the edges returned and the edges set as a standard. [19] PFOM is given by:

$$R = \frac{1}{Max(N_I, N_A)} \sum_{k=1}^{N_A} \frac{1}{1 + md^2(k)}$$

Where: N_I is the number of edges present in the image set as reference

N_A is the number of edges in the image being assessed

m is the scaling constant and d(k) denotes the distance from the actual edge to the corresponding detected edge

Figure 12. *Image taken for the purpose of the experiment [20]*

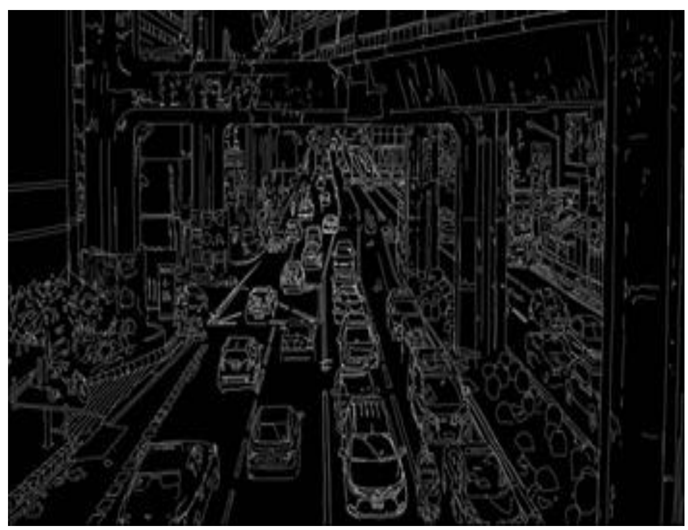

Figure 13. *Manually annotated edges to be used as ground truth for PFOM calculation*

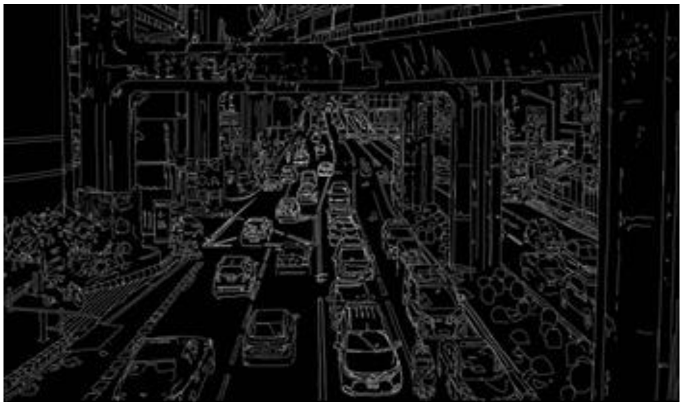

Figure 14. *Row 1 – (From Left to Right) Sobel Edge, Prewitt Edge. Row 2 – (From Left to Right) Robert, Canny Edge. Threshold [0.05 – 0.1250]*

Figure 15. *Edge detection using HED*

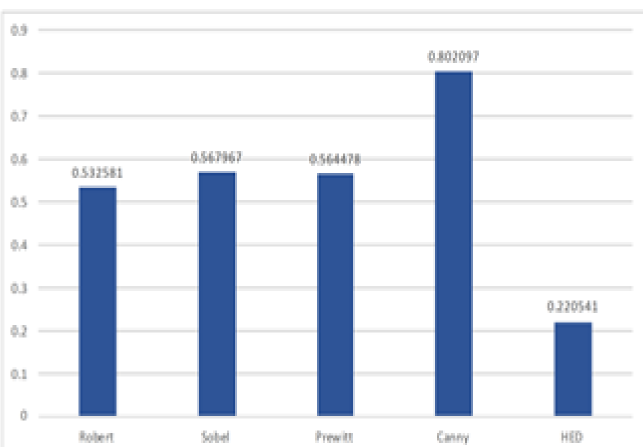

Figure 16*. PFOM of each edge detection algorithm*

5.CONCLUSION

The paper shows how important it is to not only identify the type of noise present in the image but also estimate its intensity in the image to efficiently remove it from the image by choosing the right smoothing filter and the correct parameter for that corresponding filter. Different noise type not only is removed best with different filters but the appropriate filter change with noise intensity as well. Results show that Gaussian Filter is, overall, the best option to remove Gaussian noise. The choice of filter for Speckle Noise Removal will be dependent on the noise intensity. Average Smoothing Filter would be a better choice for high noise intensity. Smoothing of Salt and Pepper noise is heavily dependent on both noise intensity and parameter taken, i.e. filter size and sigma value. Knowledge of noise intensity will dictate our choice of filter with the most appropriate parameter. HED gave bold outlines for strong edges, and was successful at distinguishing objects boundaries from edges. In case of autonomous cars, object boundaries are more important to be recognized, which HED does successfully, but still fares poorly in PFOM because the ground truth taken contains both object boundaries and edges. To assess HED better, an alternative approach at quantifying the quality of edges needs to be considered.

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